MIL-S-46874(MI)
14 December 1973
SUPERSEDING
MIS-15734
12 January 1966

MILITARY SPECIFICATION

SEAL, RF AND PRESSURE

This specification is approved for use by all activities of the Department of the Army.

1. SCOPE

1.1 Scope. This specification covers the requirements for strips of material consisting of knitted Monel metal bonded to closed cell chloroprene rubber, which functions both as an RF and pressure seal.

1.2 Classification. The strips are classified by dash numbers in accordance with cross-sectional dimensions shown in Figure 1 and Table I.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation of bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-N-281	Nickel-Copper-Alloy (Monel and R-Monel) bars, Plates, Rods, Sheets, Strips, Wire, Forgings, and Structural and Special Shaped Sections

Military

MMM-A-1617	Adhesive, Rubber Base, General Purpose

FSC 5330

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MIL-R-6130	Rubber, Cellular, Chemically Blown

STANDARDS

Federal

FED-STD-601	Rubber, Sampling and Testing

Military

MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and other publications required by suppliers in connection with specific procurement functions should be obtained from the procurement activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction sample. When specified by the procuring activity (see 6.2), a preproduction sample shall be prepared using the same methods proposed for subsequent production, and shall be subjected to preproduction inspection (see 4.3).

3.2 General. The material shall consist of knitted Monel wire strips bonded to strips of cellular rubber.

3.3 Metal.

3.3.1 Wire. The wire shall be Monel conforming with the requirements of QQ-N-281, Class A, and shall have a diameter of .0045 inch.

3.3.2 Compression of wire. Upon application of 20 plus or minus 0.5 pounds per square inch (psi) load, the knitted wire shall compress 17 to 25 percent when tested in accordance with 4.5.1.

3.4 Rubber. The rubber shall conform with the requirements of MIL-R-6130, Type II, Grade A, Condition Medium.

3.5 Adhesive.

3.5.1 Conformance. The adhesive used to bond the wire to the rubber shall conform with the requirements of MMM-A-1617, Type II.

3.5.2 Thickness. The thickness of the adhesive film between the rubber and the metal shall be .010 inch.

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3.5.3 Bond. The bond between the rubber and metal shall pass the test specified in 4.5.2.

3.6 Shape, dimensions, and tolerances.

3.6.1 Shape and dimensions.

3.6.1.1 Specifications. The material shall have the cross-sectional shape and dimensions outlined in Figure 1 and Table I as specified for each dash number.

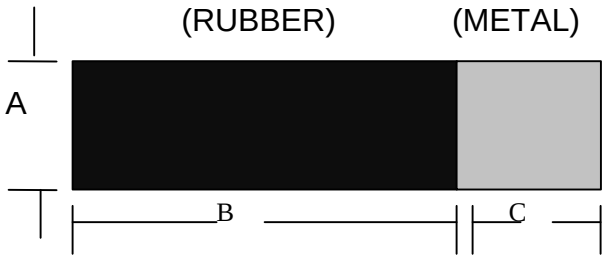

Figure 1

Table I. Dimensions

Specification Number	Dimension, Inch A	B	C
MIL-S-46874-1		.25	.25
MIL-S-46874-2		.50	.25
MIL-S-46874-3		.375	.125
MIL-S-46874-4		.50	.50
MIL-S-46874-5	.125	.50	.125
MIL-S-46874-6		.25	.125
MIL-S-46874-7		.19	.125
MIL-S-46874-8		.19	.19
MIL-S-46874-9		.125	.125
MIL-S-46874-10		.75	.125

3.6.1.2 Length. The length shall be as specified in the procurement document.

3.6.2 Tolerances. Dimensional tolerances shall be in accordance with Table II.

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Table II. Tolerances

Dimension (Inch)	Tolerance (Inch) Rubber	Metal
.06 through .125	+.03, - .015	+.015, -0
Over .125 through .19	±.03	+.015, -0
Over .19 through .375	±.03	+.03, -0
Over .375 through .50	±.06	+.05, -0
Over .50 through .75	±.06	--

3.7 Product identification.

3.7.1 Information to be shown. Unless otherwise specified in a procuring document, each separate sheet and strip of material shall have a tag attached showing the following:

a. MIL-S-46874 and appropriate dash number.

b. Vendor's name and code.

c. Manufacturer's material designation

d. Date of manufacture (year and quarter).

3.7.2 Marking. Marking shall be in accordance with the requirements of MIL-STD-129.

3.8 Workmanship. The material shall be of high quality and uniformity and shall be free of contamination, unevenness, tears, cuts, abnormal holes, and other defects. Workmanship shall be such as to ensure a product which is in conformance with this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

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4.2 Classification and tests. The testing of the material shall be classified as follows:

4.2.1 Preproduction inspection (4.3).

4.2.2 Quality conformance inspection (4.4).

4.3 Preproduction inspection. Preproduction inspection shall consist of all examinations and tests specified herein and shall be conducted only on preproduction samples. The preproduction samples shall be prepared using the same methods to be used for the manufacture of all subsequent production material.

4.4 Quality conformance inspection.

4.4.1 Inspections. Quality conformance inspection tests shall be the examination and tests required for each production lot. A production lot shall consist of finished material made on any single day from individual lots or rubber and metal components.

4.4.2 Sampling. Two 2 foot samples of each dash number material shall be taken at random from each shipment. Tests shall be run in duplicate using on specimen from each of the selected samples.

4.4.3 Quality conformance tests. Quality conformance tests shall include checking for dimensional requirements in accordance with 3.6 and comparison of the material with an approved sample to determine similarity of appearance and adequate workmanship in accordance with 3.9.

4.5 Test methods.

4.5.1 Metal compression. In order to determine conformance with 3.3.2, any conventional method may be used.

4.5.2 Adhesion. In order to determine conformance with 3.5.3, the bond between the rubber and metal shall be tested in accordance with Method 8031 or FED-STD-601. Rupture of the rubber and no adhesive failure shall be regarded as a satisfactory bond.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.

5.1.1 Level C. Unless otherwise specified in 6.2, each cross-sectional size of material shall be packed separately. The material shall not be folded or compressed. Packaging shall conform to good commercial practice and shall protect the material from contamination, damage, and change of shape during shipment and storage and shall assure safe delivery and acceptancy at destination.

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5.2 Marking. Marking shall be in accordance with MIL-STD-129 and shall include but shall not necessarily be limited to the following:

a. MIL-S-46874.

b. Vendor's name.

c. Manufacturer's material designation.

d. Quantity and dimensions.

e. Contract or procurement document number.

6. NOTES

6.1 Intended use. This material is intended for use as a compressible gasketing where both RF and pressure sealing are required.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether preproduction samples and test reports are required.

c. Special packaging and packing instructions, if required.

d. Other special instructions.

6.3 Standard test method. Due to lack of a standard test method for determining RF energy attenuation properties there are no such requirements in this specification. It is therefore recommended that the material be tested in the equipment where it is intended to be used, to assure that it is satisfactory.

6.4 Recommendation. It is recommended that the rubber component of the product be examined three years after cure date. If the rubber meets the compression deflection requirements and shows no evidence of cracks on folding, it is suggested that it be considered satisfactory for use.

6.5 Supersession data. This specification includes the requirements of Missile Interim Specification MIS-15734, dated 12 January 1966.

Custodian:
ARMY-MI

Preparing Activity:
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